Code No: 138DU **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year II Semester Examinations, June - 2024
NEURAL NETWORKS AND DEEP LEARNING
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

Used Papers 2024

PART – A

(25 Marks)

1.a)	What are the different models of ANN?	[2]
b)	Explain how supervised learning works?	[3]
c)	What do you mean by unsupervised learning?	[2]
d)	What is a competitive network?	[3]
e)	What do you mean by gradient-based learning?	[2]
f)	Give the applications of feed forward neural networks.	[3]
g)	What do you mean by regularization?	[2]
h)	What is early stopping?	[3]
i)	What do you mean by vanishing and exploding gradients?	[2]
j)	What is meta-optimization?	[3]

PART – B

(50 Marks)

2.a) Make comparison between adaptive linear element and perceptron.
b) Explain the working of back propagation algorithm. [5+5]

OR

3. What is an associative memory? Explain bidirectional associative memory architecture and algorithm for learning and give the limitations of the same. [10]

4.a) Explain the architecture of Maxnet and the algorithm for learning.
b) Explain the working of self-organizing maps. [5+5]

OR

5. What is a counter-propagation network? Explain the architectures of two types of counter-propagation networks. [10]

6. Explain the architecture, functions and importance of a feed forward deep network. [10]

OR

7.a) Explain the history, evolution and growth of deep learning.
b) Explain how to choose the type of hidden unit to use in the hidden layers of the model. [5+5]

8.a) What is L^2 regularization? When should one use L^2 regularization?
b) Explain how noise can be used as a regularization strategy? [5+5]

OR

9.a) Explain bagging and ensemble methods.
b) Explain adversarial training. [5+5]

10.a) What is the role of parameter initialization? Give an overview of various parameter initialization strategies.
b) Explain stochastic gradient descent algorithm for optimization. [5+5]

OR

11.a) Explain AdaGrad algorithm for adapting learning rates.
b) Give the applications of large scale deep learning. [5+5]

---ooOoo---

Used Papers 2024